

Tanmay Saxena

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Summary

Full Stack Engineer with 3+ years of experience building production-grade distributed systems, real-time platforms, and AI-powered applications. Delivered end-to-end features across React/React Native frontends, Node.js/Go/Python backends, and cloud infrastructure (AWS, GCP, Firebase). Strong track record of measurable performance improvements and shipping user-facing products at scale.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go, Java, SQL
Frontend: React 19, React Native (Expo), Next.js 16 (App Router, SSR), Tailwind CSS, Radix UI, Three.js
Backend: Node.js, Express.js, FastAPI, Django, WebSocket protocols, REST/GraphQL APIs
Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, CloudFormation), GCP, Firebase, Docker, GitHub Actions, CI/CD
Databases: PostgreSQL (pgvector), MongoDB, MySQL, Redis, Prisma ORM, SQLAlchemy 2.0, Alembic
AI/ML Tools: Anthropic Claude API, OpenAI GPT-4o, Google Gemini, MediaPipe, Deepgram STT, ElevenLabs TTS

Experience

Q IT Technologies

June 2025 – Present

Lead Software Developer

Gainesville, FL

- Built a real-time AI dance coaching platform with <150ms end-to-end visual feedback — MediaPipe Pose runs client-side at 30–60 FPS, transmitting only 165 floats/frame over WebSocket; video never leaves the device (privacy-by-design).
- Engineered a bidirectional voice coaching loop (Deepgram STT → Claude Haiku intent classification → ElevenLabs TTS) achieving full voice-to-response round trip under 500ms with 528+ pre-cached correction phrases.
- Designed a typed WebSocket protocol with 33+ discriminated message types covering real-time pose frames, staged feedback, voice commands, TTS audio, and full session lifecycle control.
- Architected a WebGPU-accelerated 3D Gaussian Splatting pipeline (GauHuman → ONNX → custom WGSL shaders) with 3-tier graceful fallback to ANNY mesh (WebGL) to Canvas 2D.
- Built a 6-stage LLM pipeline (Gemini 2.5 Pro) generating 80–120 page strategic intelligence reports from 50–100 aggregated video signals at \$0.40–0.60 per brand report.
- Developed a multi-platform content ingestion system processing 100–300 videos per brand from YouTube, TikTok, and Instagram with automated engagement velocity and viral potential scoring.
- Improved React Native app performance by 55% with efficient data loading, skeleton screens, FlashList, and memoized components; architected real-time Firebase communication reducing message latency by 60%.
- Engineered a Generative AI receipt processing feature integrating Vision OCR and GPT-4, cutting settlement time by over 80%.

Association for Computing Machinery (ACM) – University of Florida

Mar 2025 – May 2025

Software Engineer

Gainesville, FL

- Optimized front-end load performance by 50% for a real-time dashboard using Vite bundling and Material UI tuning.

Applyin.co

Jun 2024 – Aug 2024

SDE Intern

Remote

- Built and optimized RESTful APIs in Node.js/Express.js, improving latency by 35%; integrated JWT auth and Google SSO, reducing onboarding time by 60%.
- Reduced first input delay by 40% using React SSR, memoization, and event handler optimization.

Tata Consultancy Services (TCS)

Jan 2021 – May 2023

Systems Engineer (Developer – General Motors)

Bengaluru, India

- Automated data processing with a Django-React application, reducing manual work by 87.5%.
- Drove db optimization for a 1TB+ PostgreSQL dataset with indexing and partitioning, improving performance by 60%.
- Implemented CI/CD pipelines with GitHub Actions, cutting deployment time by 50%.

Projects

UniBazaar — Golang, PostgreSQL, React, Docker, AWS (EC2, RDS, S3, Lambda), Azure

2024

- Designed a highly-available, auto-scaling WebSocket marketplace backend using Golang and AWS, maintaining 99.9% uptime and reducing API latency by 35%.

Education

University of Florida

Aug 2023 – May 2025

M.S. in Computer Science (GPA: 3.90/4.00)

Gainesville, FL

SRM Institute of Science and Technology

Jun 2016 – May 2020

B.Tech in Electronics and Communications Engineering (GPA: 8.58/10.00)

Chennai, India